DEMO-002 - mixed STEM

Q1 $u = x^2$, $du = 2x\,dx$

$I = [\exp(u)]_0^1 = e - 1$

Q2 $\mu = 0.20$ $N = mg \cos 30$ deg

$a = g(\sin 30 \text{ deg} - \mu \cos 30 \text{ deg}) = 3.20$

$v = \mathrm{sqrt}(2as) = 4.38$ m/s

Q3 If $A^T A x = 0$, then $Ax = 0$

because A^T cancels A.

Q4

```
m = max(xs)
z = [exp(x-m) for x in xs]
return [v / sum(z) for v in z]
```